## Statistical Mechanics (MP461) Assignment 3

*Third assignment - hand in by 09.00 Friday - November 6th 2026. All* **starred** $*$ *questions to be* **attempted**. *Late assignments will not be accepted. If you have questions about this assignment, please ask your lecturer:*
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### 1. Probability of microstates

A system at constant volume $V$ and constant number of particles $N$ is in thermal contact with a heat reservoir which remains at constant temperature $T$.

(a) Derive that the probability $P(s)$ of a microstate $s$ of the system with energy $E(s)$ is given by

$$P(s) = \frac{e^{-\beta E(s)}}{Z}$$

where $\beta = \frac{1}{kT}$ and $Z$ does not depend on the state $s$.

(b) Derive an expression for $Z$.

### *2. Internal energy and heat capacity

Use the previous probability distribution to write down the expression for the internal energy $U = \langle E \rangle$ and then
(a) Show that it can be written as

$$U = -\frac{d \ln Z}{d\beta}.$$

(b) Derive an expression for the heat capacity at constant volume $C_V$ in terms of $k_B$, $T$ and $Z$.

### *3. Equi-spaced energies

Find the average energy $U = \langle E \rangle$ for an $n$-state system, in which a given state can have energy $0, \epsilon, 2\epsilon, ....n\epsilon$.

### *4. Equi-spaced energies to infinity

A harmonic oscillator, in which a given state can have energy $0, \epsilon, 2\epsilon, ....$ but this time with no upper bound.
What is the average energy $U = \langle E \rangle$ in this case?

**5. Room Temperature**

Estimate $k_B T$ at room temperature, and convert this energy into electronvolts (eV). Using this result, answer the following:
(a) Would you expect hydrogen atoms to be ionized at room temperature? (The binding energy of an electron in a hydrogen atom is 13.6 eV.)

(b) Would you expect the rotational energy levels of diatomic molecules to be excited at room temperature? (It costs about $10^{-4}$eV to promote such a system to an excited rotational energy level.)

**6. Adiabatic atmosphere**

In the last assignment, we derived the barometric equation

$$T\frac{dp}{p} = -\frac{mg}{k_B}dz \tag{1}$$

but we assumed a constant temperature of T. A better approximation, although still an approximation, is to assume that each parcel of air does **not** exchange energy with its surroundings. This means that if a parcel of air rises, it expands adiabatically.

(a) Use the fact that $p^{1-\gamma}T^{\gamma}$ is a constant to show that

$$(1-\gamma)\frac{dp}{p} + \gamma\frac{dT}{T} = 0 \tag{2}$$

and then use this together with the barometric equation (1) to estimate the so-called **adiabatic lapse rate** $dT/dz$, the rate of the drop-off in temperature with height. You may assume an atmosphere of only Nitrogen.
(b) How does your answer compare with the measured value of about $-7\frac{K}{km}$

**7. Maximum Efficiency**

What is the maximum possible efficiency of an engine operating between two thermal reservoirs, one at $100°C$ and the other at $0°C$?

***8. The Otto Cycle**

An internal combustion engine (e.g. a car engine) has an idealized description as a heat engine. A single cycle of such an engine consists of four steps.

1. A mixture of air and fuel vapour, at initial temperature $T_1$ is adiabatically compressed (in a cylinder) from an initial volume $V_1$ to a final volume $V_2 < V_1$.

2. The mixture is ignited, which raises its temperature to $T_2$, at constant volume. We can think of $T_2$ as the temperature of the hot reservoir (even though there is no hot reservoir, strictly speaking). We also ignore the chemical changes in the gas.

3. The mixture is allowed to expand adiabatically until it reaches the initial volume $V_1$

4. Finally, the gas is replaced by a new fuel/air mixture at the initial temperature $T_1$ and the same volume $V_1$. This can be modelled as if the system was simply cooling down at constant volume, in contact with a cold reservoir at temperature $T_1$.

This cycle is called the Otto cycle, after Nikolaus August Otto (1832-1891), inventor of the four-stroke engine.

a. Draw the idealized cycle in a $(p, V)$ diagram.

b. Assume that throughout the process, the gas in the cylinder always has a constant specific heat $c_V$ and energy $U = c_V nT$. Calculate the change in temperature in each of the four steps of the cycle in terms of $T_1$, $T_2$, $V_1$ and $V_2$.

c. Express the efficiency of this cycle in terms of the compression ratio $\frac{V_1}{V_2}$ and $\gamma$.

***9. Identities for partial derivatives**

In this exercise, you will derive a number of identities for derivatives which are very useful in thermodynamic calculations. Let $A$, $B$ and $C$ be three thermodynamic variables related by a single equation of state so that we can choose two of the variables as the independent variables and then the other will be a function of the first two. For example, we can choose $A$ and $B$ to be independent and then $C = C(A, B)$ is a function of $A$ and $B$, but we could also choose $A$ and $C$ or $B$ and $C$ as the independent variables. The canonical example would have $A$, $B$ and $C$ equal to the pressure, temperature and volume of a gas, but the equations below will work for any triple of quantities.

a. We take $A$ and $B$ as independent quantities. In some processes, these are varied by infinitesimal amounts $dA$ and $dB$. Write an equation for $dC$ in terms of $dA$, $dB$ and derivatives of $C(A, B)$.

b. Derive the following three identities

$$\begin{aligned}
\left(\frac{\partial C}{\partial A}\right)_B \left(\frac{\partial A}{\partial B}\right)_C + \left(\frac{\partial C}{\partial B}\right)_A &= 0 \\
\left(\frac{\partial A}{\partial B}\right)_C \left(\frac{\partial B}{\partial A}\right)_C &= 1 \\
\left(\frac{\partial A}{\partial B}\right)_C \left(\frac{\partial B}{\partial C}\right)_A \left(\frac{\partial C}{\partial A}\right)_B &= -1
\end{aligned}$$

***Hint:*** *One way to do this is to use the result of part* ***a.*** *for the case of a process in which $C$ is constant.*